

Linear Algebra Preliminary Exam

January 10, 2026

1. Let $n \geq 2$ and V be an n -dimensional real vector space, V' its dual space and $f_1, \dots, f_n \in V'$. Show that $f_1, \dots, f_n$ are linearly dependent if and only if there exists nonzero $v \in V$ such that $f_i(v) = 0$ for all $i = 1, \dots, n$.
2. Let $T : V \rightarrow V$ be a linear map where V is a linear vector space of dimension $n \geq 2$ over $\mathbb{R}$, and suppose that for some vector $v \in V$, the n vectors $v, Tv, \dots, T^{n-1}v$ are linearly independent. Show that T cannot be diagonalizable unless it has n distinct eigenvalues.
3. Let A be a complex $m \times n$ matrix and B be a complex $n \times m$ matrix where m, n are two positive integers. Suppose that μ is a nonzero eigenvalue of AB .
 - (a) Show that μ is also an eigenvalue of BA .
 - (b) Show that AB and BA have the same algebraic multiplicity for the eigenvalue μ .

4. Let $A, B \in M_n(\mathbb{C})$ be normal matrices. Show that the spectral radius ρ obeys

$$\rho(AB) \leq \rho(A)\rho(B).$$

5. Let A, B be two $n \times n$ complex matrices such that $A > 0$, i.e. A is self-adjoint and positive definite. Show that $B = 0$ if $AB + BA = 0$.
6. Let $n \geq 2$ and $A = (a_{ij})_{n \times n}$ be a complex matrix such that $A^{n-1} = 0$. Suppose n differentiable functions $f_1, \dots, f_n : \mathbb{R} \rightarrow \mathbb{C}$ satisfy for each $1 \leq i \leq n$,

$$\frac{d}{dt}f_i = \sum_{j=1}^n a_{ij}f_j.$$

Show that these functions must be linearly dependent.